

Ground Outpainting Method 讲稿（中英文对照）

1.

2.

this is whole result at the current stage.

the first one is the result of the basic seam stitching.

second one is the result of the ground outpainting .

third one is the final.

but today i want to talk is about the pipline that how i outpainting the ground. And some problems that i met now.

3. Ground Outpainting Temporal Real-Pixel Reprojection

开场 / Opening

中文：这一页讲的是 ground outpainting 的核心流程。这里的 outpainting 不是生成式补图，而是 temporal real-pixel reprojection：从同一段 AV2 log 的其他时间点借真实地面像素，投回当前 panorama。

English: The main idea is. It is temporal real-pixel reprojection: we borrow real ground pixels from other moments in the same AV2 log and project them back into the current panorama.

1. Target missing ground pixels

中文：第一步是找到当前 ERP 底部、nadir、以及 ego body 附近需要补的地面像素。我们只处理这些缺失地面区域，不是对整张 panorama 随机补图。

English: First, we identify the missing ground pixels near the ERP bottom, nadir, and ego-body occluded regions. We only target these missing ground areas, not the whole panorama.

2. ERP pixel -> 3D ground point

中文：第二步把每个缺失的 ERP pixel 反投影成一条 3D ray，再和 LiDAR-corrected ground height 相交，得到真实世界里的一个 3D ground point。

English: Second, each missing ERP pixel is back-projected into a 3D ray, then intersected with the LiDAR-corrected ground height to recover a real 3D ground point.

3. Whole-log frame search

中文：第三步是在整段 AV2 log 里找哪些过去或未来帧可能看过这个 ground point。我们不是用固定前后帧，而是按 ego displacement 选帧，因为车辆移动后才可能看到原来被车身挡住的地面。

English: Third, we search the whole AV2 log for past or future frames that may have observed this ground point. We do not use a fixed time window; we select frames by ego displacement, because the ground becomes visible only after the vehicle moves enough.

3. Whole-log frame search

中文：第三步是在整段 AV2 log 里找哪些过去或未来帧可能看过这个 ground point。我们不是用固定前后帧，而是按 ego displacement 选帧，因为车辆移动后才可能看到原来被车身挡住的地面。具体来说，我们先做 whole-log coverage search，找所有可能看到这个 hidden ground point 的视角；然后按车辆位移距离分桶，比如约 5 到 58 米，保证有足够 parallax；最后在每个距离 bucket 里选时间最近的候选帧，减少光照变化、定位误差和场景 drift。

English: Third, we search the whole AV2 log for past or future frames that may have observed this ground point. We do not use a fixed time window; we select frames by ego displacement, because the ground becomes visible only after the vehicle moves enough. Concretely, we first run a whole-log coverage search to find viewpoints that may see the hidden ground point. Then we bucket candidates by ego displacement, roughly from 5 to 58 meters, to ensure useful parallax. Finally, within each distance bucket, we keep the time-nearest candidate frames to reduce lighting changes, pose error, and scene drift.

4. Visibility-gated source sampling

中文：第四步把这个 3D ground point 投影到候选 source cameras 里，并做可见性检查。只有在 FOV 内、没有被 ego body 或动态物体遮挡的真实 source pixels 才会被保留。

English: And then, we project the 3D ground point into candidate source cameras and apply visibility checks. We keep only real source pixels that are inside the FOV and not occluded by the ego body or dynamic objects.

5. Consensus -> render or abstain

中文：最后一步是多源一致性判断。如果多个 source pixels 对同一个 ground point 的颜色一致，我们就把最可信的真实像素渲染回当前 panorama；如果证据不足或不一致，就 abstain，输出 mask 或 flat plate。

English: Finally, we run multi-source judges. If multiple source pixels agree for the same ground point, we render the most reliable real pixel back into the current panorama. If the evidence is missing or inconsistent, we abstain and output a mask or flat plate.

4. video 1

Result video slide 讲稿

中文：这一页展示的是目前 ground outpainting 做成连续视频后的结果。我们没有只挑单张好看的图片，而是每个场景连续渲染 93 帧，然后选了 4 个不同类型的场景来观察稳定性：包括比较干净的城市道路、复杂路口、crowd / 动态物体更多的场景，以及 highway 场景。

English: This slide shows our current ground outpainting results as continuous videos. For each scene, we rendered 93 consecutive frames, and selected four different scene types to evaluate stability

可以不说

中文：这里我主要想展示两点。第一，车头和车尾附近原来比较明显的 ego vehicle artifact 已经被大部分移除，地面区域可以连续地补出来。第二，在连续视频里，lane markings 和大块路面结构整体是比较连贯的，说明 temporal reprojection 在很多区域确实能找到可用的真实地面像素。

English: There are two main points here. First, the obvious ego-vehicle artifacts near the front and rear are mostly removed, and the missing ground area can be filled continuously. Second, in the video, the lane markings and large road structures are generally coherent, which suggests that temporal reprojection can recover useful real ground pixels in many regions.

中文：当然，这一页不是想说问题已经完全解决。我们也可以看到在一些局部位置，尤其是 near-nadir、强反光、动态物体遮挡、或者 source views 不一致的地方，还会出现 smear、white blob、局部模糊或者颜色不稳定。

English: At the same time, We also have same problem here. We can still see local artifacts, especially near the nadir, under strong reflections, around dynamic-object occlusions, or when source views disagree. These can appear as smearing, white blobs, local blur, or unstable color. These are the problems remained to be solved.

5. video 2

中文：这一页是没有做 ground outpainting 的 baseline video，只使用我们的多相机 ERP 拼接算法：把 AV2 ring cameras 按标定投影到 panorama，再通过接缝选择和 blending 得到连续 360 视图；可以看到基础拼接已经比较稳定，但车底和 nadir 区域仍然是黑洞/ego body artifact，所以才需要后面的 ground outpainting。

English: This is the baseline video without ground outpainting, using only our multi-camera ERP stitching pipeline: AV2 ring cameras are projected into the panorama with calibration, then seam selection and blending produce a continuous 360 view. The stitching is already stable, but the bottom/nadir region still contains black holes and ego-body artifacts, which motivates the ground outpainting step.

中文一句话：

我们的拼接算法不是简单把 7 个相机贴在一起，而是用 LiDAR 深度和相机标定做 virtual-centre 几何重投影，用 per-camera shutter pose 修正曝光时间差，再对动态物体强制 one body · one camera · one time，最终用单源接缝合成得到更稳定、少 ghost 的 360 ERP panorama。

English one sentence:

Our stitching algorithm does not simply paste the 7 cameras together; it uses LiDAR depth and camera calibration for virtual-centre geometric reprojection, corrects exposure-time offsets with per-camera shutter poses, and enforces one body, one camera, one time for dynamic objects, then uses single-source seam compositing to produce a more stable, less-ghosted 360 ERP panorama.

How to present this slide / 整页怎么串起来讲

中文：这一页讲的是我们最初拼接算法的核心思路：remove the cause, not just hide the artifact。多相机拼接的主要问题来自三类不一致：空间不一致、时间不一致、动态物体不一致。

English: This slide explains the core idea of our original stitching algorithm: remove the cause, not just hide the artifact. The main problems in multi-camera stitching come from three types of inconsistency: spatial inconsistency, temporal inconsistency, and dynamic-object inconsistency.

中文：第一，7 个 AV2 cameras 不是同一个光心，所以如果直接把所有图像投到 ego origin，近处物体和路面结构会有 parallax 错位。Virtual centre 这一步用相机内外参和 LiDAR 深度，把每个 camera pixel 按真实 3D 几何重投影到 panorama，使静态结构在接缝处更对齐。

English: First, the 7 AV2 cameras do not share the same optical center. If we directly project all images from the ego origin, nearby objects and road structures will have parallax misalignment. The virtual centre step uses camera intrinsics, extrinsics, and LiDAR depth to reproject each camera pixel according to real 3D geometry, so static structures align better across seams.

中文：第二，7 个相机不是完全同一时刻曝光，最多有几十毫秒 shutter offset。车辆在这段时间里已经移动，所以我们给每个 camera 使用它自己曝光时刻的 ego pose，通过 SE(3) interpolation 得到 per-camera shutter pose。这样静态世界不会因为时间差产生错位。

English: Second, the 7 cameras are not exposed at exactly the same time; they can have tens of milliseconds of shutter offset. The ego vehicle moves during that interval, so each camera must use the ego pose at its own exposure time. We obtain this per-camera shutter pose through SE(3) interpolation. This prevents static-world misalignment caused by timing differences.

中文：第三，动态物体不能像背景一样跨相机混合。对于车辆、行人这些 moving objects，我们强制 one body, one camera, one time。也就是说，一个物体整体从一个相机、一个时间点取出来，再通过 mask、3D box association、局部 registration 和 DP seam 放回 panorama。这样可以避免 ghost car 和 seam-cut car。

English: Third, dynamic objects cannot be mixed across cameras like the static background. For moving objects such as cars and pedestrians, we enforce one body, one camera, one time. In other words, each object is rendered as a whole from one camera at one exposure time, then placed back into the panorama using masks, 3D box association, local registration, and a DP seam. This helps avoid ghosted cars and seam-cut cars.

中文：所以这页的 takeaway 是：拼接算法不是简单地把 7 个 camera 贴到一起，而是先修正空间几何，再修正曝光时间，最后对动态物体做单源一致性处理。这样得到的 baseline panorama 在水平 360 视野上会比较稳定；但由于 AV2 没有真正向上和向下看的相机，所以 top / bottom / nadir 的黑洞仍然需要后面的 ground outpainting 来处理。

English: So the takeaway is: the stitching algorithm is not simply pasting the 7 cameras together. It first corrects spatial geometry, then corrects shutter-time misalignment, and finally handles dynamic objects with single-source consistency. This gives a stable horizontal 360-degree baseline panorama, while the top, bottom, and nadir gaps still need the later ground outpainting step.

Baseline Stitching Algorithm 备忘录（中英文对照）

一句话版本 / One-sentence version

中文：这一页展示的是没有 ground outpainting 的 baseline stitching video：我们把 AV2 的 7 个 ring cameras 按标定投影到 ERP panorama，用 seam selection / single-source compositing 做拼接，所以中间水平视野比较稳定，但顶部和底部因为没有相机覆盖，仍然会保留黑洞和 ego body artifact。

English: This slide shows the baseline stitching video without ground outpainting: we project the 7 AV2 ring cameras into an ERP panorama using calibration, then stitch them with seam selection and single-source compositing. The horizontal view is mostly stable, but the top and bottom remain black or ego-body artifacts because they are not covered by the cameras.

1. 输入是什么 / What are the inputs?

中文：输入是 AV2 的 7 个 ring camera 图像，以及每个相机的内参、外参和相机之间的标定关系。这些相机围绕车顶安装，不是同一个光心，所以它们看到的画面会有 parallax。

English: The inputs are the 7 AV2 ring camera images, together with each camera's intrinsics, extrinsics, and calibration. These cameras are mounted around the vehicle and do not share the same optical center, so their views contain parallax.

2. 为什么要投影到 ERP / Why project to ERP?

中文：我们的目标是生成一个 360 度 panorama，所以使用 ERP，也就是 equirectangular projection。ERP 的横轴表示 360 度 yaw，纵轴表示上下视角。每个 ERP pixel 对应一个空间方向。

English: Our target is a 360-degree panorama, so we use ERP, or equirectangular projection. The horizontal axis represents 360-degree yaw, and the vertical axis represents pitch. Each ERP pixel corresponds to a direction in 3D space.

3. 每个 camera 怎么放到 panorama 上 / How is each camera placed into the panorama?

中文：对每个 ERP pixel，我们根据它对应的 3D viewing direction，判断这个方向会落在哪些 source cameras 的 FOV 里。然后用相机标定把这个方向投影回原始 camera image，采样真实像素。

English: For each ERP pixel, we use its 3D viewing direction to check which source cameras can see it. Then, using camera calibration, we project that direction back into the original camera image and sample the real pixel.

4. 多个相机重叠时怎么选 / How do we handle overlapping cameras?

中文：相邻相机之间会有重叠区域。如果简单 alpha blending，把两个相机的像素平均，就容易出现 ghosting，因为两个相机的光心不同，同一个近处物体会在两个图里位置不一样。所以我们的 baseline 更偏向 single-source stitching：每个位置尽量只选一个相机的像素，而不是把两个几何不对齐的像素混在一起。

English: Adjacent cameras have overlapping regions. If we simply alpha-blend two cameras, ghosting can appear because the cameras have different optical centers, so nearby objects appear at different positions. Our baseline therefore uses single-source stitching: each output pixel is mostly taken from one selected camera, instead of blending two geometrically misaligned pixels.

5. seam selection 是什么 / What is seam selection?

中文：seam selection 的作用是在相机重叠区域里决定“切线”走哪里。理想情况下，seam 应该避开近处车辆、行人、车道线、路沿等容易产生错位结构，尽量走在纹理平滑、两个相机看起来更一致的位置。

English: Seam selection decides where the boundary between cameras should pass inside overlapping regions. Ideally, the seam should avoid nearby cars, pedestrians, lane markings, curbs, and other structures that are sensitive to parallax, and instead pass through smoother areas where the two camera views agree better.

6. 为什么还有黑色区域 / Why are there still black regions?

中文：因为 AV2 的 ring cameras 主要看水平方向，并没有真正覆盖天空顶部和车底正下方。所以 baseline stitching 只能拼接相机真实看到的区域；没有相机覆盖的 top / bottom / nadir 区域就会保持黑色，车身附近也会留下 ego body artifact。

English: The AV2 ring cameras mainly cover the horizontal field of view. They do not fully observe the sky top or the area directly under the vehicle. Therefore, baseline stitching can only stitch regions that were actually seen by the cameras; uncovered top, bottom, and nadir regions remain black, and ego-body artifacts remain near the vehicle.

7. 和 ground outpainting 的关系 / Relation to ground outpainting

中文：baseline stitching 解决的是“多相机怎么拼成一个 360 panorama”的问题；ground outpainting 解决的是“底部和车身附近没有相机直接看到的地面怎么补”的问题。所以这一页展示的是 outpainting 之前的结果，它说明基础拼接已经稳定，但也暴露了底部黑洞和 ego artifact 需要后续处理。

English: Baseline stitching solves the problem of combining multiple cameras into one 360-degree panorama. Ground outpainting solves a different problem: how to fill the bottom and ego-adjacent ground area that no current camera directly observes. This slide shows the result before outpainting: the basic stitching is stable, but the bottom black holes and ego artifacts still need to be handled later.

如果被问到“为什么不直接 blend?” / If asked: why not just blend?

中文：因为这些相机不是同一个光心，近处物体在不同相机里的位置不一致。直接 blend 会把两个错位的物体平均在一起，产生半透明重影。为了保持清晰和真实，我们更倾向于 hard select / single-source compositing。

English: Because the cameras do not share the same optical center, nearby objects appear at different positions in different cameras. Direct blending averages two misaligned observations and creates translucent ghosts. To keep the result sharp and faithful, we prefer hard selection or single-source compositing.

如果被问到“baseline 已经解决了什么?” / If asked: what does the baseline already solve?

中文：baseline 已经能把 7 个 ring cameras 拼成一个连续的水平 360 ERP，并且大部分建筑、车辆、道路结构在视频里是稳定的。它主要没有解决的是 top / bottom FOV gap、nadir black hole、ego body artifact，以及 near-ground seam 的局部几何错位。

English: The baseline can already stitch the 7 ring cameras into a continuous horizontal 360-degree ERP, and most buildings, vehicles, and road structures are stable in the video. What it does not solve are the top/bottom FOV gaps, the nadir black hole, eg